HOMEWORK #5

Spring 2026
EIN 4360: Facilities Planning & Material Handling

Instructor: **Nicolas Grimaldi**
Due: **Sunday, April 5th at 11:59 PM**

Lecture 10: Schedule Design II

Problem 1. Gordon Ramsay has tasked you with determining how many cooking stations (pans) to assign to each sous-chef in his restaurant. The activity cart below (Fig. 1.1) shows the chef (O-1) and the pan they are using to cook the food (M-1). Times are shown on the left side of the figure, and each step is shown on the key to the right.

a) What is the ideal assignment (n')?
b) If the sous-chefs' hourly rate is $30/hr and the cost of each additional pan is $2.5/hr. How many pans should be assigned per sous-chef?

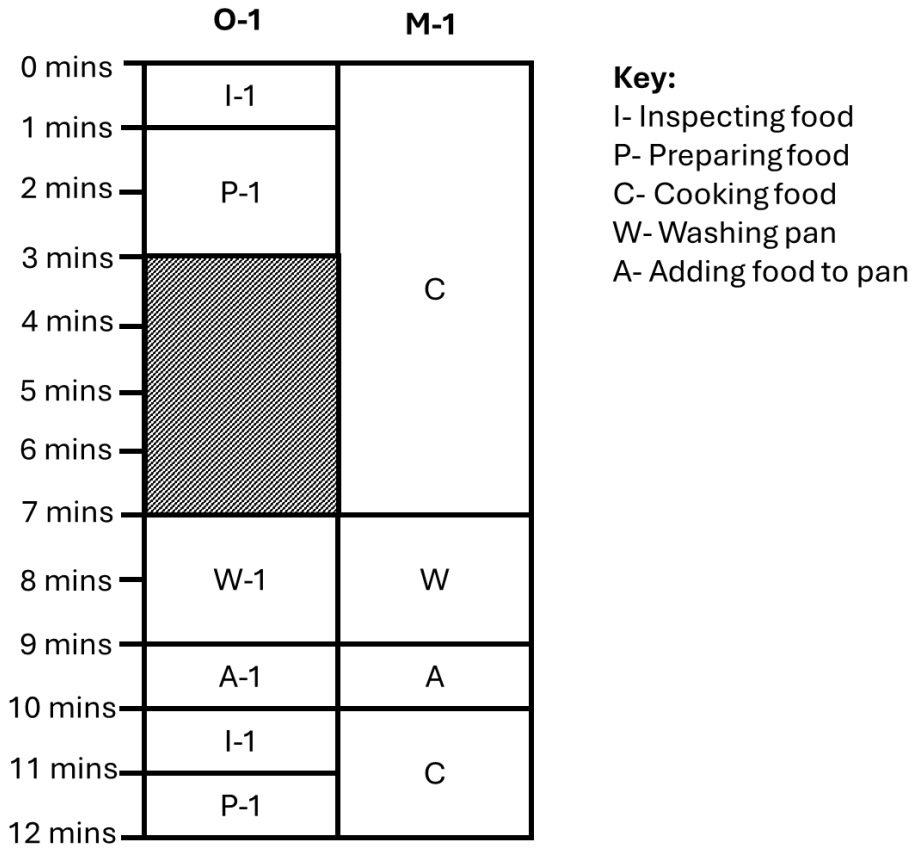

Fig. 1.1. Activity chart for Gordon's sous-chefs.

Problem 2. Suppose your family is planning on hosting a barbeque for some family members. It is expected that 12 family members will attend the barbeque, and your family needs your help determining how many burgers they should prepare for the barbeque. Historically, not all burgers scheduled became edible, some were burnt or dropped. Based on previous barbeques, you determine the following probabilities (Table 2.1). Each burger costs $5 to make, on average, even if a burger is burnt or a patty is dropped, the buns and toppings can still be used in the future, so $2 is salvaged back. If 12 good burgers are made, each family member is willing to pay back $8 per burger to help offset costs. However, if less than 12 good burgers are made, everyone agreed to order pizza instead, in this case, the burgers scheduled can still be salvaged for $2 each. If more than 12 good burgers are made, everyone will still pay $8 for each of the 12 burgers, and the remaining good burgers will be salvaged for $2 each.

a) How many burgers should we schedule to maximize the expected profit?
b) If this many burgers are scheduled, what is the probability of losing money?

Table 2.1. Estimated probability distribution.

# of Good Burgers	Number of Burgers Scheduled				
	12	**13**	**14**	**15**	**16**
10	0.3	0.1			
11	0.4	0.3	0.2	0.1	
12	0.3	0.4	0.3	0.2	0.1
13		0.2	0.4	0.4	0.3
≥14			0.1	0.3	0.6

Lecture 11: Break Even Analysis

Problem 3. You are tasked with deciding the next investment for your company. One option involves purchasing a new robot to help around the facility. This robot is expected to improve productivity and cashflow by 10% every year for the next ten years. The cost of the robot is $50,000, and it can be salvaged at the end of the ten years for $7,500. Alternatively, a treasury bond could be purchased, which has a fixed 5% annual interest rate and can be cashed out at the end of the ten years. Your expected/forecasted cash flow currently (without considering the robot) is shown in Table 3.1. Given this information determine:

a) The return on investment (ROI)
b) The payback period (PBP)
c) The net present value (NPV)

Table 3.1. Expected cashflow for the company.

Year	Expected Cashflow
1	$45,000
2	$50,000
3	$60,000
4	$60,000
5	$75,000
6	$80,000
7	$85,000
8	$90,000
9	$90,000
10	$100,000